United International University
Department of Computer Science and Engineering
CSE 4531: Computer Security
Final Examination: Spring 2026
Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions. The number to the right of each question denotes its marks.

1(a) Alam wants to send the message M=4 to Samir using the RSA cryptosystem. Samir chooses two prime numbers p=3 and q=11. Furthermore, [6]

- Samir computes his public key (n,e) as (33, 7) and sends it to Alam.
- Alam uses this public key to encrypt the message and sends the ciphertext to Samir.
- Samir decrypts the ciphertext using his private key.

Your tasks are given as follows:

a) **Compute** the ciphertext C from Alam's encryption.

b) **Compute** the decrypted message M′ from Samir's decryption.

c) **Draw** a diagram showing the exchange of keys and messages between Alam and Samir.

1(b) Suppose Host A and Host B are using a platform to setup secure session between them, and they apply the **Diffie-Hellman key exchange protocol** with the following parameters: [4]

- Public parameters: modulus p = 13, base g = 11
- Private key of A: 10, Private key of B: 9

Generate the shared secret key between Host A and Host B.

2(a) Suppose an attacker wants to hijack an active TCP connection (between Alice and Bob) but is not in the same subnetwork of Alice or Bob. What challenges would the attacker face in carrying out this attack? With the help of a diagram, **explain** how the attack could be performed despite these challenges. [5]

2(b) With the help of a diagram, **explain** how a stack-based buffer overflow attack overwrites a function's return address. **Explain** how stack canaries can help prevent buffer overflow attacks. [5]

3(a) Considering the network scenario shown in Fig. 1, answer the following queries: [3+3]

i) **Explain** how the ARP (Address Resolution Protocol) can be exploited to perform a Man-In-The-Middle (MITM) attack by the malicious device against both victims (V1 and V2).

ii) **Find out** the resulting ARP tables of V1, V2 and Router after the MITM attack:

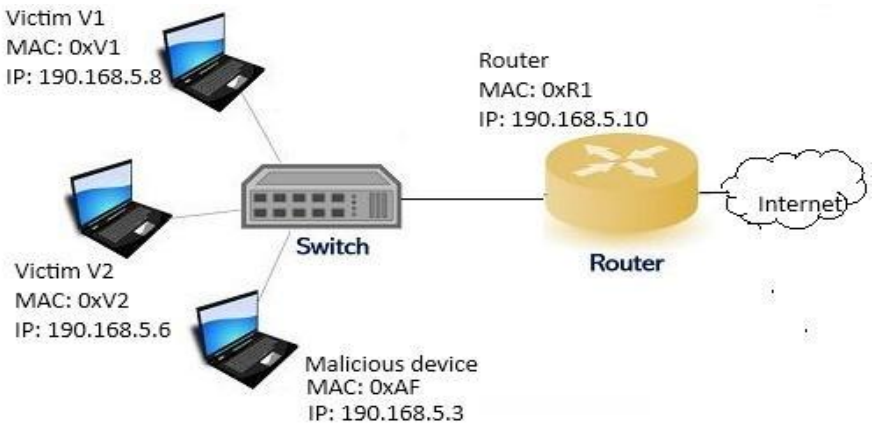


Fig 1: A Network Topology

3(b) A ransomware attack locks all medical records, thereby preventing doctors from accessing patient data during an emergency. Which part of the CIA Triad is violated? Suggest methods to defend against such attacks. [4]

4(a) A company employee logs into the organization's payroll portal and keeps the browser open. Later, the employee visits an unrelated website that contains hidden code. Without the employee's knowledge, the website automatically sends a request to the payroll portal to change the employee's direct-deposit bank account information. Because the employee is already authenticated, the payroll system accepts the request and updates the account details, even though the employee never intended to make the change.
What type of web attack is being **demonstrated** in this scenario, and why is it successful? Also, **mention** some prevention strategies to stop this type of attack. [5]

4(b) Alice installed an unverified mobile application that streams various TV channels in high definition without any payment. After some days, she experienced multiple unrecognized activities in her Facebook account, such as running ads, adding payment information, and so on. She also discovered that her Facebook account has been linked to an Instagram account that does not belong to her. **Identify** what kind of malware infected her mobile phone with a proper explanation. Also, **mention** some prevention strategies to stop this type of attack. [5]